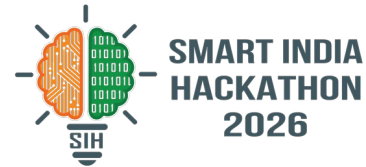
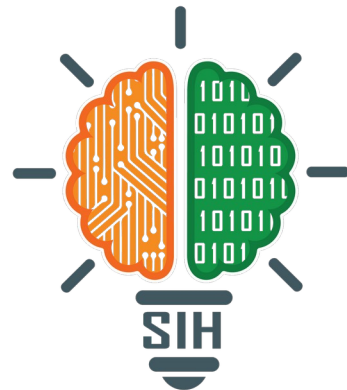


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26097
- Problem Statement Title – AI-driven voice assistant for livelihood mapping and NSQF-aligned skilling recommendations for SC communities under the GIA component of PM-AJAY
- Theme – Agriculture, FoodTech & Rural Development
- PS Category – Software
- Team ID – _____ (fill from the SIH portal)
- Team Name (Registered on portal) – Cognify



SMART INDIA HACKATHON 2026

Proposed solution · one free phone call that turns a person's skills into the right work



Wrong course, then no job

that is what happens to SC families today (SIH26097 · MoSJE)

PM-AJAY pays for **skill training** and **small-business support** (up to ₹50,000). But people are picked and placed through **paper forms** and **field visits** — those who can't read, or don't know the trades, land in the wrong course.

41% got a job

after government training

Source: CAG audit 2025

43% have no phone

of rural women, of their own

Source: NFHS-6, 2023-24

34% cannot read

of SC adults

Source: Census 2011

WHO WE SERVE

20.14 crore SC people → **≈ 5.9 crore** eligible (below poverty line) → **≈ 50,000** get GIA support each year → **Less than 1 in 1,000** a year → every seat must fit

Census 2011 (via MoSJE Handbook 2021) · SC poverty ratios 31.5% rural / 21.7% urban (Planning Commission, 2011-12) · GIA: 2,01,006 beneficiaries in 2021-25 → ≈ 50,000 a year (PIB, 19 Mar 2025)

ONE CALL about 10 minutes, on any keypad phone — four steps



Missed call → free call-back

within a minute, in his language

He gives a missed call; SkillCall calls back from a **govt 1600 number** in the language he picks on the keypad.
No app, no internet, no reading.



Keys for facts, voice for skills

so a speech error can never hurt

Age, education, travel distance are **pressed on keys** — eligibility never depends on speech. Then "Aap kya kaam karte ho?" — the AI maps his work to an **official NCO code** and asks about his skills.



Confirm → 1-4 options that fit

every fact checked before it is spoken

He hears everything back and presses 1 to confirm — **the AI writes the profile, no paper**. Then only courses, schemes or jobs near him that truly fit. **He can ask any doubt.**



Apply & follow-up

CSC → officer → 3 calls

SMS with **Case ID + nearest CSC**; the officer approves in the AJAY app; SkillCall calls at **1, 3 and 6 months** until he is earning.

WHAT MAKES IT DIFFERENT five things a paper form can never do



Smart questions

each depends on the last answer

Bike mechanic → engine, tyres, oil — **not a generic form**.



Profile writes itself

officers only review and approve

The AI fills the **full profile table** in the proper format.



Verified facts only

govt data + SATYA double-check

Answers come only from loaded rules, NQR courses, NCO codes, centres; SATYA matches each fact to a DB row — **no match → not said**.



Cheapest AI first

bigger model, then a human, only if needed

Common answers get **cached** — cost falls over time.



Web portal too

same case, no call needed

Mobile + OTP login for the family, CSC or officer to view or edit.

Also answers the 5 GIA issues in the brief: planning · beneficiary selection · placement after training · coordination · too few ground staff

TECH STACK *Indian, open where possible*

Calls & voice

Exotel / Plivo

WhatsApp API

Pipecat voice server

AI models

Saaras v3 • STT

Bulbul v3 • TTS

Sarvam-30B / 105B

IndicConformer

Bhashini

SATYA (ours)

Backend & data

Python

FastAPI

Celery

Redis

PostgreSQL

pgvector

Apps & govt links

React

Leaflet maps

NCO / NQR

AJAY app*

DigiLocker*

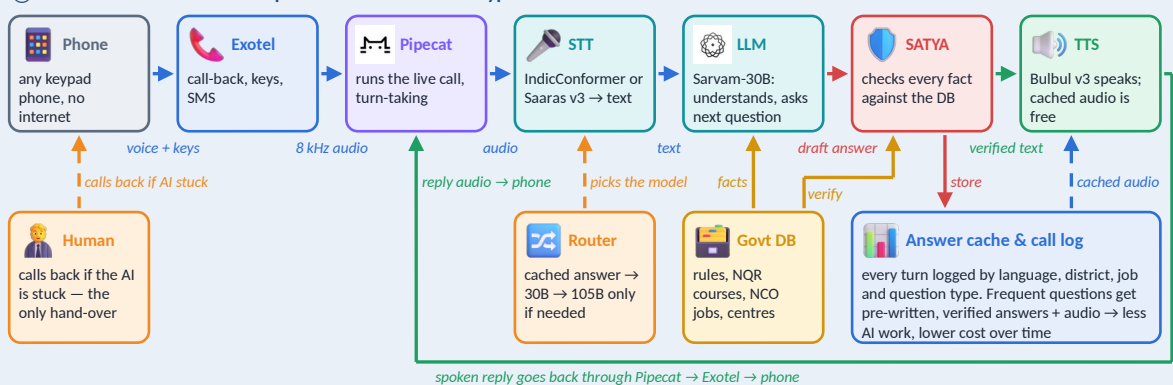
Docker • MeghRaj*

* needs a government MoU.

[Full architecture, API docs and code](#) → [GitHub repo](#) (add link)

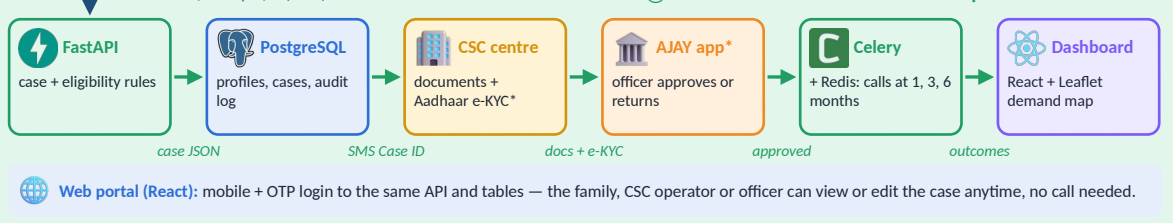
HOW IT WORKS *two lanes: during the call (AI, in seconds) and after it (people, in days)*

① DURING THE CALL — each spoken turn = 1.2-2.2 s • keypad turns are instant



call ends → confirmed profile (JSON) → FastAPI

② AFTER THE CALL — the confirmed profile becomes a case



AI understands → verified data decides → SATYA checks → humans approve

Data rules: no Aadhaar number stored · raw audio deleted after confirmation · consent logged (DPDP Rules 2025) · every edit in the audit log

Can it be built and run? **Yes** — every part already runs in India, the prototype can be built today, and one completed call costs about ₹25–54.

ALREADY WORKING IN INDIA *nothing here is untested*



10 languages out of 22

are already in use

UIDAI already runs Sarvam's voice AI for Aadhaar on its own servers, in 10 of Sarvam's 22 languages.

Source: PIB / MeitY 2025



12 lakh calls a day

already handled by a govt line

Kilkari, the government health-advice line, runs this volume every single day.

Source: BMJ Global Health 2021



≤ 20% word error

on real Indian phone calls

Best Indian speech AI, tested on **36,691 speakers** in 15 languages (536 hours).

Source: Voice of India 2026 (arXiv)



1.2–2.2 s per reply

fast enough to feel like a talk

End of speech 0.5 s + speech-to-text 0.1–0.3 s + AI first word 0.3–1 s + voice 0.2 s + network 0.1–0.2 s.

Source: Sarvam docs • NVIDIA 2026

ONE COMPLETED CALL COSTS ₹25–54 *rates per minute → one call of ≈ 16 min*

Cost item	Rate	Per call
Phone line (Exotel)	₹0.80–1.80 per minute (+ unfinished calls, 30-s billing)	₹14.7–33.7
Speech → text (Saaras v3)	₹0.50 per minute, only his speech (50–60% of the call)	₹4.2–5.1
Text → speech (Bulbul v3)	₹3 per 1,000 characters; fixed questions pre-recorded	₹2.4–6.0
AI brain (Sarvam-30B / 105B)	~90% of turns on 30B; 105B only when needed	₹0.25–0.5
SMS + human call-back	4–6 SMS; 10–20% of callers need a human for 5–6 min	₹2.2–5.3
Hosting & helpline number	servers, database, number — spread per call	₹1.0–3.0
ONE COMPLETED CALL	all-in ≈ ₹1.6–3.4 per minute	₹25–54

Per 1,000 missed calls, at the average = ₹40 a call: 420–517 people get a complete, verified profile for ≈ ₹17,000–21,000 in total.

50% are reached on the 1st call and 76% by the 3rd; ~80% of those finish the interview and ~85% of those finish the options. Only 10–20% (≈ 40–100 people) need a human counsellor call. **Estimated** — every step is worked in the calculations sheet.

Prices & funnel: Sarvam API pricing • Indian telephony rate cards 2026 • Kilkari (BMJ 2021) • funnel = our estimate

WHAT COULD GO WRONG — AND WHAT ALREADY HANDLES IT

✗ Village dialects are hard to understand ✓ **solved:** facts by keypad, every answer read back, bigger model, then a human
✗ People ignore unknown numbers ✓ **solved:** he calls first; the call-back comes from an official 1600 number

THE GOVERNMENT CAN AFFORD IT *we add ~1% to what is already spent*



0.05–0.10% of the budget is all SkillCall needs

All of India: **₹1.1–2.1 crore a year** vs PM-AJAY's ₹2,140 crore (2026–27).



Only 1% more per person and the other 99% stops being wasted

The govt already spends **₹60,700** on one beneficiary. SkillCall costs **₹220–420** more — **under 1%** — to make sure that ₹60,700 goes to the right course.



₹9–13 lakh to start a 6-month pilot in one block

≈ 1,000 families: build, **test with real users**, then scale to the state.

Cost falls further over time. Frequent questions get pre-written, verified answers and audio (answer cache, slide 3); Sarvam bills cached input at ₹10.98 vs ₹29.28 per 1M tokens; at national scale ~70% of speech moves to free IndicConformer → **₹21–49 per profile.**

✗ The AI could say something wrong ✓ **solved:** SATYA blocks anything not in the government database
✗ Slow replies bore callers ✓ **solved:** streaming voice, pre-recorded questions, keypad turns are instant

Who gains? SC families, CSC operators, district officers and the Ministry. Every number shows how it was worked out.

WHAT CHANGES *four numbers, each with its working*



41% → 70% placed

the gap we close

Only 41% of trained people get a job today; 70% is the govt's own target. Per 1,000 trainees, even 50% means 90 more people working.

Source: CAG 2025 · MSDE norms



62% → 73% stay in the job

when told the truth before joining

+11 percentage points (men) in a field trial of two honest job-information sessions. SkillCall gives that on the call.

Source: J-PAL · Chakravorty et al. 2024



30 h → 5 h a month

of officer time, per 40 cases

Today 45 min each (interview + 3 follow-up calls); with SkillCall a 7.5-min review. India: $50,000 \times 37.5 \text{ min} = 31,250 \text{ h a year} \approx 18 \text{ full-time officers}$.

Our estimate: AJAY app's data entry not counted



₹12.5–15 cr a year

not wasted on wrong matches

If wrong matches fall in 5 of every 100 cases: $2,500 \times ₹50,000\text{--}60,700 \text{ support} = 6\text{--}14 \times \text{SkillCall's yearly cost}$.

Our estimate: PIB 2025 average support

1–1.5 lakh profiles a year so every one of the $\approx 50,000$ GIA seats goes to the person it fits — 2–3 applicants screened per seat (calculations sheet \$1.9)

WHO WINS *and what each one actually gets*



SC family

right course, first time

Free, in his own language — and calls until he is earning.



CSC operator

the case arrives pre-filled

Only documents and Aadhaar e-KYC are left to do.



District officer

ready, checked cases

In the AJAY app, plus a demand map by block for the yearly plan.



Ministry

one record, call to job

Each outcome marked confirmed (employer / EPFO*) or self-reported.

BENEFITS *social · economic · governance · environment*



Social

Women and non-readers included; right course the first time.



Economic

Less money on wrong training; more people earning — proved by follow-up.



Governance

One shared case for CSC, officer, partner and bank; planning data, not paper.



Environment

No paper forms, fewer trips to offices, runs on the shared govt cloud.



Government data

PIB, Rajya Sabha reply (19 Mar 2025) — GIA: 2,01,006 beneficiaries, ₹1,219.80 cr pib.gov.in

PIB PM-AJAY explainer (May 2025) — eligibility, 5% admin & MIS line pib.gov.in

CAG Report No. 20 of 2025 — PMKVY: 41% placed, no skill-gap analysis cag.gov.in

NFHS-6 (2023-24) — women's phone ownership 63.6%, rural 57.4% rchiips.org

Census 2011 (MoSJE Handbook 2021) · SC poverty ratios 2011-12 (Lok Sabha) socialjustice.gov.in



Technology

Sarvam API pricing & docs — Saaras v3, Bulbul v3, 30B / 105B; voice-agent guide docs.sarvam.ai

NVIDIA × Sarvam 2026 — Sarvam-30B serving, first token < 1 s (P95) developer.nvidia.com

Artificial Analysis — Sarvam-30B ≈ \$0.03 per 1M tokens (blended) artificialanalysis.ai

AI4Bharat IndicConformer · Pipecat · Bhashini ai4bharat.iitm.ac.in



Research

Chakravorty et al. 2024 / J-PAL — honest job info: 62% → 73% stay in job povertyactionlab.org

Bashingwa et al. 2021, BMJ Global Health — Kilkari call answer rates gh.bmj.com

Voice of India 2026 — speech-AI benchmark on real Indian phone calls arxiv.org

Medhi et al. 2011, ACM TOCHI — voice interfaces for low-literacy users dl.acm.org



Standards & policy

PM-AJAY portal & AJAY app launch (26 May 2026) — PIB pib.gov.in

NCO-2015 occupation codes · NQR / NCVET job roles nqr.gov.in

DPDP Rules 2025 — consent and data protection meity.gov.in

TRAI 1600-series rules for government calls (2026) traf.gov.in

MeitY (Lok Sabha 2023) — 5.58 lakh CSCs, 4.44 lakh at gram panchayats eparlib.nic.in

SEE MORE *code, prototype, video and every calculation — one click each*



GitHub repo

code, architecture docs, API reference

[\(add link\) →](#)



Live prototype

try a SkillCall conversation in the browser

[\(add link\) →](#)



Demo video

2-minute walkthrough of a real call

[\(add link\) →](#)



Calculations sheet

every number, its source and full working

[\(add link\) →](#)